

Andhra Pradesh – Grade 10

2018 – Mathematics

1. Find the HCF of 60 and 100 by using Euclid division lemma.

Solution:

$$100 = 60 \times 1 + 40$$

$$60 = 40 \times 1 + 20$$

$$40 = 20 \times 2 + 0$$

Hence, HCF = 20

2. Write $A = \{3, 9, 27, 81\}$ in set-builder form.

Solution:

$$A = \{x : x = 3^n, n \text{ belongs to } \mathbb{N} \text{ \& } n < 5\}$$

3. Find the value of k for which the pair of equations $2x + ky + 3 = 0$, $4x + 6y - 5 = 0$ represent parallel lines.

Solution:

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$$\frac{2}{4} = \frac{k}{6} \neq \frac{3}{-5}$$

$$k = 3 \text{ or } k \neq \frac{-18}{5}$$

4. Find the volume of right circular cone with radius 3 cm. and height 14 cm.

Solution:

$$\text{Volume} = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \times \frac{22}{7} \times 3 \times 3 \times 14$$

$$= 132 \text{ cm}^3$$

5. Find the zeroes of the polynomial $x^2 - 3$ and verify the relationship between the zeroes and the coefficients.

Solution:

$$\text{If } x^2 - 3 = 0$$

$$x = \pm\sqrt{3}$$

$$\text{L.H.S} = x^2 - 3$$

$$= (\sqrt{3})^2 - 3 = 0 = \text{R.H.S verified}$$

6. How many three digit numbers are divisible by 3?

Solution:

First three digit No is 100

First divisible Number = 102

Last Number = 999

Common difference = 3

Series is 102, 105, 108, 999

$$a_n = a_0 + (n-1)d$$

$$999 = 102 + (n-1) 3$$

$$\frac{897}{3} = n - 1$$

$$299 + 1 = n$$

$$n = 300$$

7. A solid iron rod has a cylindrical shape. Its height is 11 cm and base diameter is 7 cm. Then find total volume of 50 rods.

Solution:

Volume of 1 cylinder = $\pi r^2 h$

$$= \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 11$$

$$= 121 \times \frac{7}{2}$$

$$\text{Total volume of 50 rods} = 121 \times \frac{7}{2} \times 50$$

$$= 847 \times 25$$

$$= 21175 \text{ cm}^3$$

8. Find the roots of $x + \frac{6}{x} = 7$, $x \neq 0$.

Solution:

$$x + \frac{6}{x} = 7$$

$$x^2 - 7x + 6 = 0$$

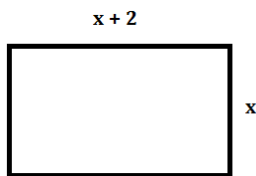
$$x^2 - 6x - x + 6 = 0$$

$$x(x - 6) - 1(x - 6) = 0$$

$$x = 6, 1$$

9. Length of rectangle is 2 units greater than its breadth. If the area of the rectangle is 120 sq. units, then finds its length.

Solution:



$$(x + 2) \times x = 120$$

$$x^2 + 2x - 120 = 0$$

$$x^2 + 12x - 10x - 120 = 0$$

$$x(x + 12) - 10(x + 12) = 0$$

$$(x + 12)(x - 10) = 0$$

$$\text{Length} = 10 + 2 = 12 \text{ cm}$$

10. (a) Hari went to a bank to withdraw Rs. 2000. He asked the cashier to give the cash Rs. 50 and Rs. 100 notes only. He got 25 notes in all. Can you tell how many notes, each of Rs. 50 and Rs. 100, he received?

(or)

(b) How many spherical balls can be made out of a solid cube of lead, whose edge measures 66 cm. and each ball being 3 cm. in radius?

Solution:

(a) let No. of 50 Rs. Notes be x

No. of 100 Rs. Notes be $25 - x$

$$50 \times x + 100(25 - x) = 2000$$

$$50x + 2500 - 100x = 2000$$

$$500 = 50x$$

$$x = 10$$

50 Rs. Note $\rightarrow 10$

100 Rs. Note $\rightarrow 15$

(b)

$$\text{Volume of cube} = 66 \times 66 \times 66$$

$$\text{Volume of spherical ball} = \frac{4}{3} \pi \times 3^3$$

$$\begin{aligned} \text{No. of ball} &= \frac{66 \times 66 \times 66 \times 7}{4 \times 22 \times 3 \times 3 \times 3} \\ &= 2541 \end{aligned}$$

11. (a) Show that $\sqrt{3}$ is irrational.

OR

(b) If $A = \{x : x \text{ is a natural number}\}$

$B = \{x : x \text{ is an even number}\}$

$C = \{x : x \text{ is an odd number}\}$

$D = \{x : x \text{ is a prime number}\}$

Then find $A \cup B$, $A \cap C$, $B \cap C$ and $B \cap D$. What do you notice?

Solution:

(a)

Let $\sqrt{3}$ be rational

$$\text{So, } \sqrt{3} = \frac{a}{b}$$

$$a = b\sqrt{3}$$

on L.H.S a is still rational

R.H.S $b\sqrt{3}$ is irrational

Which contains the equality

Hence, $\sqrt{3}$ is irrational.

11. (b)

$$A \cup B = \{\text{All natural number}\}$$

$$A \cap C = \{\text{All odd number}\}$$

$$B \cap C = \{ \}$$

$$B \cap D = \{2\}$$

12. (a) The sum of the reciprocals of Rahman's age, (in years) 3 years ago and 5 years from now is $\frac{1}{3}$. Find his present age.

OR

(b) If the sum of first 7 terms and 15 terms of an A.P. are 98 and 390 respectively, then find the sum of first 10 terms.

Solution:

12(a) Let x

$$@ \text{ 3 years ago} = x - 3$$

$$@ \text{ 5 years from} = x + 5$$

$$\frac{1}{x+3} + \frac{1}{x-3} = \frac{1}{3}$$

$$3(x+5+x-3) = (x-3)(x+5)$$

$$6x+6 = x^2+5x-3x-15$$

$$0 = x^2-4x-21$$

$$0 = x^2-7x+3x-21$$

$$x(x-7)+3(x-7)$$

$$x=7 \text{ and } x=-3$$

So, present age = 7 years

12(b)

$$\text{Sum} = \frac{n}{2}(2a + (n-1)d)$$

$$98 = \frac{7}{2}(2a + (7-1)d)$$

$$28 = 2a + 6d$$

$$14 = a + 3d \quad \dots\dots(1)$$

$$390 = \frac{15}{2}(2a + (15-1)d)$$

$$\frac{780}{15} = 2a + 14d$$

$$26 = a + 7d \dots\dots(2)$$

Solving (1) and (2)

$$14 = a + 3d$$

$$26 = a + 7d$$

$$\underline{-12 = -4d}$$

$$d = 3$$

$$a = 14 - 3d = 14 - 3 \times 3 \\ = 5$$

@ 10 terms

$$\begin{aligned} \text{Sum} &= \frac{10}{2}(2a + (n-1)d) \\ &= 5(2 \times 5 + 9 \times 3) \\ &= 5(10 + 27) \\ &= 37 \times 5 = 185 \end{aligned}$$

13. (a) Solve the quadratic polynomial $p(x) = x^2 - x - 6$ by graphical method.

(OR)

(b) The perimeter of a rectangular plot is 32m. If the length is increased by 2 m. and the breadth is decreased by 1 m, the area of the plot remains the same. Find the length and breadth of the plot. (Use graph)

Solution:

13. (a)

$$y = x^2 - x - 6$$

y intercept @ $x = 0$

$$y = 0 - 0 - 6 = -6$$

x intercept @ $y = 0$

$$x^2 - x - 6 = 0$$

$$x^2 - 3x + 2x - 6 = 0$$

$$x(x-3) + 2(x-3)$$

$$(x-3)(x+2)$$

$$x = 3 \text{ and } x = -2$$

13. (b)

$$l \rightarrow l + 2 = 16 - b - 2 = 18 - b$$

$$b \rightarrow b - 1$$

$$\text{perimeter} \rightarrow 32$$

$$2 \times (l + b) = 32$$

$$l + b = 16 \quad \rightarrow l = 16 - b$$

Area is constant

$$(18 - b)(b - 1) = lb = (16 - b)b$$

$$18b - 18 - b^2 + b = 16b - b^2$$

$$19b - 18 = 16b$$

$$3b = 18$$

$$b = 6$$

$$l = 16 - b$$

$$= 16 - 6 = 10 \text{ cm}$$

14. Which of the following is a terminating decimal?

Options:

(a) $\frac{10}{81}$

(b) $\frac{41}{75}$

(c) $\frac{8}{125}$

(d) $\frac{3}{14}$

Answer:(c)

15. The value of $\log_2 32 = \dots$

(a) 2

(b) 32

(c) $\frac{1}{5}$

(d) 5

Answer(d)

Solution:

$$\frac{\log 32}{\log 2} = \frac{5 \log 2}{\log 2} = 5$$

16. If '3' is one of the zeroes of $p(x) = x^2 + kx - 9$, then the value of k =

(a) 0

- (b) 1
- (c) 2
- (d) 3

Answer(a)

Solution:

$$3^2 + k \times 3 - 9 = 0$$

$$3k = 0$$

$$k = 0$$

17. The pair of equations $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are consistent, then....

- (a) $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$
- (b) $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$
- (c) $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$
- (d) a and c

Answer(d)

18. If α, β, γ are the zeroes of the cubic polynomial $ax^3 + bx^2 + cx + d$ and ($a \neq 0$), then $\alpha\beta\gamma = \dots\dots\dots$

- (a) $\frac{d}{a}$
- (b) $-\frac{d}{a}$
- (c) $-\frac{b}{a}$
- (d) $\frac{c}{a}$

Answer (b)

19. Which term of A.P, 18, 15, 12, equals to '0'?

- (a) 4
- (b) 5
- (c) 6
- (d) 7

Answer(d)

Solution:

$$0 = 18 + (n-1) \cdot -3$$

$$18 = 3n - 3$$

$$\frac{21}{3} = n = 7$$

20. If $A \subset B$, $n(A) = 4$ and $n(B) = 6$, then $n(A \cup B) = \dots\dots$

- (a) 10
- (b) 6
- (c) 4
- (d) 2

Answer(c)

Solution:

If $A \subset B$, then $A \cup B = B$

$$n(B) = 6$$

So, options (c)

21. If $k, 2k+1, 2k+3$ are three consecutive terms in A.P., then find the value of k .

- (a) 1
- (b) 0
- (c) 2
- (d) 3

Answer(a)

Solution:

$$2(2k+1) = k + 2k + 3$$

$$4k + 2 = 3k + 3$$

$$k = 1$$

22. A quadratic polynomial, whose zeroes are $\sqrt{2}$ and $-\sqrt{2}$ is

- (a) $x^2 - 4$
- (b) $x^2 + 4$
- (c) $x^2 - 2$
- (d) $x^2 + 2$

Answer: (c)

Solution:

$$x^2 - 2$$

$$\text{Sum of root} = \sqrt{2} - \sqrt{2} = 0$$

$$\text{Product} = \sqrt{2} \times -\sqrt{2} = -2$$

$$x^2 - 0x - 2 = 0$$

$$x^2 - 2 = 0$$

23. If $a_n = \frac{n}{n+1}$, then $a_{2017} = \dots\dots\dots$

- (a) $\frac{2017}{2016}$
- (b) $\frac{2017}{2018}$
- (c) $\frac{2017}{2019}$
- (d) $\frac{2018}{2017}$

Answer(b)

24. A cylinder and cone have bases of equal radii and are of equal heights, then their volumes are in the ratio.....

- (a) 1 : 1
- (b) 1 : 3
- (c) 3 : 1
- (d) 1 : 9

Answer(c)

Solution:

$$\frac{\text{volume of cylinder}}{\text{volume of cone}} = \frac{\pi r^2 h}{\frac{1}{3} \pi r^2 h} = 3:1$$

25. Total surface area of a solid hemisphere of radius 7 cm. is....cm².

- (a) 2π
- (b) 49π
- (c) 147π
- (d) 98π

Answer: (c)

Solutions: $T.S.A = 3\pi r^2$
 $= 3 \times \pi \times 7^2$
 $= 147\pi$

26. A quadratic equation $ax^2 + bx + c = 0$ has two distinct real roots,if.....

- (a) $b^2 - 4ac > 0$
- (b) $b^2 - 4ac < 0$
- (c) $b^2 - 4ac = 0$
- (d) None of the given

Answer: (a)

27. The degree of the polynomial $5x^7 - 6x^5 + 7x - 4$ is

- (a) 5

- (b) 6
- (c) 7
- (d) 4

Answer: (c)

28. n^{th} terms of a progression $a, ar, ar^2, \dots$ is.....

- (a) $a r$
- (b) $a r^2$
- (c) $a+(n-1)r$
- (d) $a r^{n-1}$

Answer: (d)

Solutions:

29. Which of the following equations has the solution of (1,-1)?

- (a) $3x - 2y = 6$
- (b) $3x + 2y = 6$
- (c) $3x - 2y = 5$
- (d) $3x + 2y = 5$

Answer: (c)

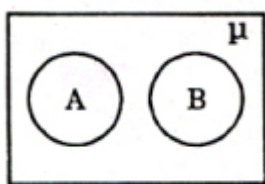
Solutions:

30. If $A = \{x : x \text{ is a letter in the word } \{\text{EXAMINATION}\}\}$, then its roster form is

- (a) $A = \{e, x, m, i, n, a, t, o, s\}$
- (b) $A = \{e, x, m, i, n, a, t, o\}$
- (c) $A = \{e, x, m, a, i, n, t, s\}$
- (d) $A = \{e, x, m, i, n, t, o\}$

Answer: (b)

31. The following Venn diagram indicates.....



- (a) $A \subset B$
- (b) $B \subset A$
- (c) A, B are disjoint sets.
- (d) $A = B$

Answer: (c)

32. The discriminant of the quadratic equation $px^2 + qx + r = 0$ is

- (a) $p^2 - 4qr$

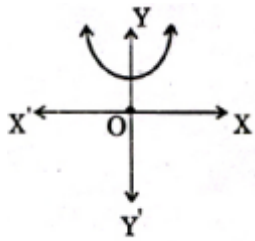
(b) $q^2 - 4pr$

(c) $q^2 + 4pr$

(d) $r^2 - 4pq$

Answer: (b)

33. Number of zeroes that can be identified by the following figure



(a) 0

(b) 1

(c) 2

(d) 3

Answer: (a)